

COMBINATORICS ASSIGNMENT 2

ROB TOMPKINS

6. Prove that

$$P(n, k) = k! \binom{n}{k}.$$

Proof. First let's consider how we choose our k -permutations of $[n]$. We begin by choosing n elements, then $n - 1$, then all the way down to $n - k + 1$, so we see that

$$P(n, k) = n(n - 1)(n - 2) \cdots (n - k + 1) = \frac{n!}{(n - k)!}.$$

On the right-hand side, we are choosing k -elements from $[n]$ in an unsorted fashion, but then we need to sort them in k ways which is $k!$, and we see that

$$k! \binom{n}{k} = k! \left(\frac{n!}{k!(n - k)!} \right) = \frac{n!}{(n - k)!}.$$

□

10. Prove that:

$$\sum_{k=0}^n r^{n-k} \binom{n}{k} = (r + 1)^n$$

Solution.

□

16. Prove Vandermonde's Identity:

$$\binom{m + n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r - k}$$

Proof. The left-hand side counts all the k -element subsets of $[m + n]$. The right-hand side does the same thing according to the number of elements chosen from $[n]$. Indeed, we can first choose i elements from $[n]$ in $\binom{n}{i}$ ways, then choose the remaining $k - i$ elements from the set $\{n + 1, n + 2, \dots, n + m\}$ in $\binom{n}{i}$ ways. □

19. Prove the identity:

$$\binom{n + 1}{r + 1} = \sum_{k=r}^n \binom{k}{r}$$

Proof. Let's go ahead and expand out the right hand side here so we see that:

$$\binom{n+1}{r+1} = \binom{n}{r} + \binom{n}{r+1} + \binom{n}{r+2} + \cdots + \binom{n}{n} = \binom{n+1}{k+1}$$

Now with regards to the identity in question, the left-hand side clearly counts all the $r+1$ -element subsets of $[n+1]$. And if we separate the right-hand side into cases according to the largest element. That is, there are $\binom{n}{r}$ subsets of $[n+1]$ that have $r+1$ elements whos largest element is $r+1$; there are $\binom{n}{r+1}$ subsets of $[n+1]$ that have $r+1$ elements whos largest element is $r+2$ and so on. In general there are $\binom{n}{r+k}$ subsets of $[n+1]$ that have $r+1$ elements whos largest element is $r+k+1$ for all $k \leq n-r$. $\square$

20. Prove the identity:

$$2^n - 1 = \sum_{k=0}^{n-1} 2^k$$

Proof. We begin by noticing that the left hand side counts the non-empty subsets of $[n]$. To see the count of the right-hand side we count by classifying all the subsets of $A \subseteq [n]$ by each's largest element $m \in A$ where $m \in \{1, 2, \dots, n\}$, $\max(A) = m$. We notice that every element of A is chosen from $[n]$, and elements $\{1, 2, \dots, m-1\}$ can be included or not independently. So we have that:

$$2^n - 1 = \sum_{m=1}^n 2^{m-1}.$$

A simple shift in indices $k = m - 1$ then results in:

$$2^n - 1 = \sum_{k=0}^{n-1} 2^k.$$

$\square$

22. Prove the identity:

$$\binom{n}{k} \binom{k}{t} = \binom{n}{t} \binom{n-t}{k-t}$$

Proof. The left-hand side is the number of ways that we can choose S of k -elements from $[n]$, and then subsequently choose t -elements from S . Regarding counting the right-hand side we see that we are immediately choosing t -elements from our original set of $[n]$. But then we are left with a set of $[n-t]$ from which we need to choose $k-t$ elements. So it works. $\square$

24. Prove the identity:

$$\sum_{k=0}^m \binom{m}{k} \binom{n}{r+k} = \binom{m+n}{m+r}$$

Solution.

$\square$

26. Prove the identity:

$$\sum_{k=0}^n k^2 \binom{n}{k} = n(n-1)2^{n-2} + n2^{n-1} = n(n+1)2^{n-2}$$

Solution.

□

33. Prove the identity:

$$\sum_{k=n-s}^{m-r} \binom{m-k}{r} \binom{n+k}{s} = \binom{m+n+1}{r+s+1}$$

Solution.

□